

UNDER HYPOTHESIS H, FORMALLY GENERIC PRIMES BLOCK THE POINTWISE SIGNED-GRAPH APPROACH TO THE ERDŐS–STRAUS CONJECTURE

ANONYMOUS

ABSTRACT. For a prime $p = 4t + 1$ the signed integer solutions of $4/p = 1/x + 1/y + 1/z$ form a finite graph, two solutions being adjacent when they share a denominator. The *seed* $(t, -2pt, -2pt)$ is available for every such p , and the *seed-component conjecture* asserts that its component always contains an all-positive solution; it would imply the Erdős–Straus conjecture for $p \equiv 1 \pmod{4}$. We explain why, conditionally on Schinzel’s Hypothesis H, this pointwise programme is blocked. Arguments that use only congruence data and the shape of factorisations fail at “formally generic” primes, and we prove precise forms of this principle. Unconditionally, we classify all escapes from the seed of length at most three, show that every prime outside Mordell’s six classes modulo 840 escapes in two moves, and bound the exceptional sets by $N(\log N)^{-11/2}$ (distance > 2) and $N(\log N)^{-10}$ (distance > 5). Under Schinzel’s Hypothesis H the seed distance is unbounded. Under prime-tuple hypotheses, moreover, the whole seed component has no positive vertex for infinitely many primes. This was obtained first, and in a stronger form, by a companion computation that assumes only Dickson’s conjecture for 159 linear forms. We report that result without re-proving it. On a subprogression, that construction also exhibits a positive solution outside the sterile component. We give an independent confirmation from Hypothesis H for 6402 explicit polynomials of degree at most two, via a certified finite closure of 7883 formal vertices. The bounded-radius result, the Elsholtz–Tao “odd-square” remark and Schinzel’s polynomial-identity theorem share one mechanism. At a formally generic prime every number the argument meets factors formally, and in a square class the Bright–Loughran character dichotomy leaves no formal solution positive. We also record further obstructions: sterile components larger than the seed component, the triviality of large p -free buckets, and parity obstructions. The Erdős–Straus conjecture itself is untouched.

1. INTRODUCTION

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

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has a solution in positive integers. It suffices to treat primes, and Mordell's identities settle all primes outside the six classes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ [13]. The conjecture has been verified numerically to 10^{18} [11]. It is open; no published argument proves it.

1.1. The signed graph and the seed. Allowing *signed* denominators changes the problem completely: for every odd $n \geq 3$,

$$\frac{4}{n} = \frac{1}{(n-1)/2} + \frac{1}{(n+1)/2} - \frac{1}{n(n^2-1)/4}$$

(an identity attributed to Jaroma, see [4, §1 and ref. 14]), and for fixed n there are only finitely many of them [4, Lemma 3.10]. Throughout, $p = 4t + 1$ is a prime.

Definition 1.1. A *vertex* is a multiset $\{x, y, z\}$ of nonzero integers with $4/p = 1/x + 1/y + 1/z$. It is *positive* if $x, y, z > 0$ and *nonpositive* otherwise. The *signed graph* G_p joins two distinct vertices when they share a denominator (as an integer). The *seed* is $\sigma = (t, -2pt, -2pt)$; its connected component is the *seed component* $C_\sigma(p)$. The *seed distance* $\text{dist}(p) \in \{0, 1, \dots\} \cup \{\infty\}$ is the graph distance from σ to the nearest positive vertex of $C_\sigma(p)$.

Conjecture 1.2 (seed-component conjecture). *For every prime $p \equiv 1 \pmod{4}$ the seed component $C_\sigma(p)$ contains a positive vertex.*

A positive vertex is an ES solution. So Conjecture 1.2 implies ES for all primes $p \equiv 1 \pmod{4}$; the primes $p \equiv 3 \pmod{4}$ are classical. Moreover Conjecture 1.2 is an *algorithmic* statement: starting from an object available for every p , walk through the finite graph by exact divisor moves until positivity is reached. It was the working hypothesis of a pointwise programme. This paper explains why that programme is blocked.

1.2. The main message. The obstruction is an instance of a principle that is informally familiar from Elsholtz–Tao. Type I and Type II solutions do not exist at odd squares [8, Prop. 1.6], so a method showing that they exist at p must fail when p is replaced by an odd square. This rules out finite sets of covering congruences [8, §1, remark after Prop. 1.6, p. 6], and polynomial identities on square classes [8, p. 8]. (ES itself holds at odd squares, e.g. $4/9 = 1/4 + 1/6 + 1/36$; the obstruction concerns the methods.) We make this precise for explorations of the signed graph.

Principle (formal blindness; informal). *Call an argument formal if, along a residue class $q \equiv q_0 \pmod{M}$ of the parameter $q = (p-1)/24$, it inspects only the values at q of finitely many fixed polynomial expressions. Every decision it takes must also depend only on the class of q and on the shape of the factorisations of these values (which irreducible polynomial factors occur, with which exponents). At a formally generic prime, where all polynomial values met are primes, such an argument sees only formal solutions. If the class is a square class, there is no formally positive solution (Schinzel;*

Bright–Loughran). So, under Hypothesis H , a formal argument confined to a square class fails for infinitely many primes.

The principle is a heuristic summary, not a theorem. Its precise content is given by the following results. Proposition 4.6 is its unconditional core: there is no formally positive vertex in a square class. Theorem 4.7 shows that a closed formal exploration of G_p with no formally positive member contains the whole seed component. Theorem 4.9 covers explorations of bounded radius from the seed, under H . The whole seed component is handled by finite closures (Theorems 4.11 and 4.13). For these, nonpositivity is verified directly rather than deduced from the square-class mechanism. The conclusion (§7) is that a pointwise proof must control the *actual* factorisations of specific shifted integers, for instance by using the size of p .

1.3. Results. Status labels follow the campaign convention: PROVED, CONDITIONAL (proved here from a named hypothesis), CERTIFIED (an exact finite computation re-checked independently), EVIDENCE (finite data), and REPORTED (a result proved elsewhere, stated here with an indication of what we re-checked). “Verified numerically” never means “proved”.

Theorem A (PROVED; §§2–3).

- (1) *A vertex is positive if and only if the two denominators of equal p -adic valuation, with p removed, have product a quadratic non-residue modulo p (Bright–Loughran [4]; Yamamoto for the positive direction).*
- (2) *Every p -free denominator outside $[1, 2t]$ lies in at most one vertex. Every Type II p -divisible denominator lies in at most two vertices, all of one sign class.*
- (3) *$\text{dist}(p) \leq 3$ if and only if one of three explicit divisor conditions holds (Theorem 3.1). Every prime $p \equiv 1 \pmod{4}$ outside Mordell’s six classes has $\text{dist}(p) = 2$.*
- (4) *$\#\{p \leq N, p \equiv 1 \pmod{4} : \text{dist}(p) > 2\} \ll N(\log N)^{-11/2}$ and $\#\{p \leq N, p \equiv 1 \pmod{4} : \text{dist}(p) > 5\} \ll N(\log N)^{-10}$.*

Theorem B (CONDITIONAL on Hypothesis H ; Theorem 4.9). *For every k there are infinitely many primes $p = 24q + 1$, with q prime, such that every vertex within distance k of the seed is nonpositive. Under the Bateman–Horn conjecture there are $\gg_k N(\log N)^{-C_k}$ of them up to N . Hence, under H , no bounded-depth search from the seed proves ES for all primes $p \equiv 1 \pmod{4}$.*

Theorem C (§4.7).

- (1) (REPORTED [1], conditional on Dickson; the stronger version. Finite packet re-checked here; eventual-closure argument not re-verified.) *There are 159 explicit linear forms $F_i(n)$, among them q and $p = 24q + 1$, with the following property. For all sufficiently large n at which every $F_i(n)$ is prime, the seed component of p has no positive vertex. On an admissible subprogression, p nevertheless has an explicit positive solution, which therefore lies outside $C_\sigma(p)$. Dickson’s*

conjecture for these 159 forms gives infinitely many sterile seed components. For the 159 forms restricted to the subprogression, it gives infinitely many such primes that also have a positive solution.

- (2) (This paper, CONDITIONAL on H; independent confirmation.) *Under Hypothesis H for an explicit family of 6402 polynomials of degree at most two, infinitely many primes $p = 24q + 1$ have a seed component with no positive vertex. That component consists of the values of 7883 explicit formal vertices.*

In particular, by part (2), which is proved here, Conjecture 1.2 is false under Hypothesis H. Part (1) reports that the same holds under Dickson’s conjecture.

The first part of Theorem C is due to the companion project [1], and it is stronger than the second part in two respects. It needs only Dickson’s conjecture, the linear case of Hypothesis H, for 159 forms. It also exhibits an ES solution at the primes of the subprogression $n \equiv 507 \pmod{857}$, so those primes are counterexamples to *reachability*, not candidates against ES. The second part was obtained independently, by a different certificate (a closure over a polynomial frame rather than an affine packet), and it was re-verified by an independent engine. The two projects share their early history (the signed graph, the seed, and the outer-anchor bounds for $z < 0$ and $z \geq p$ of Theorem 2.10); they separated before either sterility result was found.

1.4. What is not claimed. ES is not proved or disproved here. Conjecture 1.2 is *open* unconditionally; no sterile seed component is known at an actual prime, and the families of Theorem C are far beyond computational reach. Numerically every prime $p \equiv 1 \pmod{4}$ below 10^{12} has $\text{dist}(p) \leq 3$ (§6). This is EVIDENCE, and by Theorem B the pattern is conditionally false.

1.5. Credits. The character dichotomy (Theorem 2.5) is due to Bright and Loughran [4, Thms. 1.2, 1.5]; its positive direction goes back to Yamamoto [17] (see [4, App. A]). Finiteness of signed solutions is [4, Lemma 3.10]. The labels and the Type I chart are Elsholtz–Tao coordinates [8, (2.1), (2.6), (2.7), (2.18), (2.21)]. Rigidity of p -divisible denominators between types was observed for positive solutions by Jiang [10, Thm. 3.2]; that preprint has since been withdrawn. The polynomial-identity obstruction is Schinzel’s [14]; see [8, p. 8]. Bright and Loughran recover the odd-square vanishing [8, Prop. 1.6] from the Brauer–Manin obstruction [4, Cor. 1.4]. The half-dimension idea in Theorem 3.6 is Dahan’s [5, Lemma 4.2, Thm. 4.3]. We found no prior source for the signed graph, the seed, Theorem 3.1, Theorem B or Theorem C(2), apart from the companion project [1]; Theorem B is best read as the H-conditional form of the Elsholtz–Tao remark, and its novelty is unconfirmed.

2. THE SIGNED GRAPH

2.1. Fibres and finiteness. Fix a denominator z of some vertex. Since p is odd, $4z \neq p$, so $4/p - 1/z = r/s$ with $\gcd(r, s) = 1$, $s > 0$ and $r \neq 0$.

Lemma 2.1 (fibres; PROVED). *The vertices containing z are exactly the triples*

$$\left(z, \frac{s+D}{r}, \frac{s+s^2/D}{r}\right),$$

where D runs over the signed divisors of s^2 with $D \neq -s$ and $r \mid s+D$. The divisors D and s^2/D give the same vertex. Since $\gcd(D, r) = 1$, integrality of the second entry implies that of the third.

Proof. $1/y + 1/w = r/s$ is equivalent to $(ry-s)(rw-s) = s^2$. Put $D = ry-s$. Then $D(s+s^2/D) = s(D+s)$, and $\gcd(D, r) = 1$ because $D \mid s^2$ and $\gcd(r, s) = 1$. The value $D = -s$ gives $y = 0$. $\square$

Every vertex has a positive denominator at most $3p/4$, since otherwise its positive terms would sum to less than $4/p$. By Lemma 2.1, G_p is therefore finite [4, Lemma 3.10].

2.2. Valuations, types and labels.

Lemma 2.2 (valuations; PROVED). *Every vertex has exactly one or exactly two denominators divisible by p , and each of them has p -adic valuation exactly 1.*

Proof. Clearing denominators gives $4xyz = p(xy+yz+zx)$, so p divides some denominator. If p divided all three, multiplying by p would write 4 as a sum of three reciprocals of nonzero integers; but such a sum is at most 3. If exactly one denominator $p^k m$ is p -divisible, then $1/(p^k m) = 4/p - 1/x - 1/y$ has valuation -1 , so $k = 1$. Suppose x is p -free and $p^i m, p^j n$ with $i, j \geq 1$, $p \nmid mn$. If $i < j$, then $1/(p^j n) = 4/p - 1/x - 1/(p^i m)$ would have valuation $\geq -i > -j$, which is impossible. So $i = j$. If $i = j \geq 2$ then $|4/p - 1/x| \leq 2/p^2$. This forces $x > 0$ and $x \leq p^2/(4p-2) < p/2$, whence $0 < |4x-p| \leq 2x/p < 1$, a contradiction. $\square$

A vertex is of *Type I* if exactly one denominator is p -divisible and of *Type II* if two are. For a p -divisible denominator pm put $\lambda(pm) = 4m - 1 \pmod p$.

Lemma 2.3 (labels; PROVED). *At a Type I vertex the p -divisible denominator has $\lambda = 0$. At a Type II vertex (x, pm, pn) ,*

$$(4m-1)(4n-1) \equiv 1, \quad n(4m-1) \equiv m \pmod p.$$

In particular both labels are nonzero, and a p -divisible denominator occurs only in vertices of one type.

Proof. Multiply the equation by p and reduce modulo p . For (x, y, pm) this gives $4 \equiv 1/m$. For (x, pm, pn) it gives $4 \equiv 1/m + 1/n$, that is, $4mn \equiv m + n$. $\square$

On positive points these labels are Elsholtz–Tao parameters. Write $a_{\text{ET}}, b_{\text{ET}}, \dots, f_{\text{ET}}$ for the coordinates of [8, §2], and drop the subscripts inside the next two displays. A positive Type I vertex is

$$(abdp, acd, bcd) \quad [8, \text{Prop. 2.2}],$$

so $m = a_{\text{ET}} b_{\text{ET}} d_{\text{ET}}$ and $4m - 1 = p e_{\text{ET}}$ is [8, (2.1)]. A positive Type II vertex is

$$(abd, acdp, bcdp) \quad [8, (2.22), \text{Prop. 2.6}].$$

There $4m - 1 = f_{\text{ET}}$ by [8, (2.18)], and the identity above is [8, (2.21)] reduced modulo p .

Lemma 2.4 (small anchor; PROVED). *Every vertex has a p -free denominator x with $1 \leq x \leq 2t$. The p -free denominator of a Type II vertex is in $[1, 2t]$.*

Proof. For Type II, (x, pm, pn) gives $p/x = 4 - 1/m - 1/n \geq 2$, so $0 < x \leq p/2$. For Type I, (x, y, pm) , the label gives $m \equiv -t \pmod{p}$, hence $m \leq -t$ or $m \geq 3t + 1$. So $R := 4/p - 1/(pm)$ is positive, with $R > 4/p$ if $m < 0$ and $R \geq 3/(3t + 1)$ if $m > 0$. The positive terms among $1/x, 1/y$ sum to at least R , so one of x, y is positive and at most $2/R$. This bound is less than $p/2$, respectively at most $2t + 2/3$. $\square$

In the positive case this is the Monks–Velingker bound [12, Thm. 2.1(i)].

2.3. The character dichotomy.

Theorem 2.5 (Bright–Loughran; PROVED in [4]). *Let v be a vertex. Let A, B be the two denominators of v with equal p -adic valuation, each divided by p^{v_p} . Then*

$$v \text{ is positive} \iff \left(\frac{AB}{p} \right) = -1.$$

Derivation from [4]. By Lemma 2.2 the valuation pattern is $(1, 0, 0)$ or $(0, 1, 1)$, so exactly one pair has equal valuation. Label the denominators u_1, u_2, u_3 so that u_2, u_3 is this pair. Then $u_2/u_3 = A/B$ is a p -adic unit and $v_p(u_1 u_3) = 1$ in both patterns. By [4, (3.1), Lemma 3.4] the Hilbert symbol $(-u_1/u_3, -u_2/u_3)_p$ equals $\left(\frac{-u_2/u_3}{p} \right) = \left(\frac{-AB}{p} \right) = \left(\frac{AB}{p} \right)$, since $p \equiv 1 \pmod{4}$. For $n = p$, [4, Thm. 1.2] says that this symbol is -1 at positive solutions, and [4, Thm. 1.5] says that it is $+1$ at integer solutions which are not positive. The symbol is invariant under permuting the u_i ([4, §1], referring to [4, Prop. 2.6]). $\square$

The direction “positive $\Rightarrow$ non-residue” is [4, Cor. 1.3], which unifies Yamamoto’s conditions [17]. By [4, Thm. 1.6], the Brauer group of the Erdős–Straus surface modulo constants is $\mathbb{Z}/2$, generated by the class behind Theorem 2.5. So this character is the *only* Brauer–Manin invariant. It identifies positivity exactly, but it cannot create it.

Corollary 2.6 (Type II homogeneity; PROVED). *A Type II vertex (x, pm, pn) is positive if and only if $\left(\frac{\lambda(pm)}{p} \right) = -1$. Hence all vertices containing a given Type II denominator pm have the same sign class.*

Proof. By Lemma 2.3, $n \equiv m\lambda(pm)^{-1}$, so $\left(\frac{mn}{p} \right) = \left(\frac{\lambda(pm)}{p} \right)$. Apply Theorem 2.5. $\square$

2.4. The Type I chart.

Lemma 2.7 (Type I chart; PROVED). *Let pm be the p -divisible denominator of a Type I vertex and x one of its p -free denominators. Write $x = t + a$; then $m = ph - t$ for an integer h . Put $H = 4h - 1$ and $e = (4a - 1)h - a$. Then:*

- (1) $e \neq 0$, $e \mid x^2$, and the third denominator is xm/e ;
- (2) $Hx = m + e$ and $(4a - 1)H - 4e = 1$, so $e \mid x^2 \iff e \mid m^2$;
- (3) conversely, if $t + a \neq 0$, $e \neq 0$ and $e \mid (t + a)^2$, then $(t + a, p(ph - t), (t + a)(ph - t)/e)$ is a Type I vertex. In particular $t + a$ is automatically p -free;
- (4) if $x \geq 1$, the vertex is positive if and only if $a \geq 1$ and $h \geq 1$.

Proof. Since $4m \equiv 1$, $m \equiv -t \pmod{p}$. Next, $4m - 1 = pH$, so $4/p - 1/(pm) = H/m$, and $1/x + 1/y = H/m$ gives $y = mx/(Hx - m)$. Expanding, $Hx - m = (4h - 1)(t + a) - (ph - t) = (4a - 1)h - a = e$. Since $\gcd(e, H) = 1$, the conditions $e \mid mx = (Hx - e)x$, $e \mid x^2$ and $e \mid (Hx - e)^2 = m^2$ are all equivalent. For (3): if $p \mid t + a$, then two denominators would be p -divisible, one of them with label $\lambda = pH \equiv 0$, contrary to Lemma 2.3. For (4): $a, h \geq 1$ give $e \geq 2ah > 0$. Conversely a positive vertex has $pm > 0$, so $h \geq 1$, and $1/x < 4/p$, so $x > p/4$, that is, $a \geq 1$. $\square$

On positive Type I points, with the ET coordinates above, $H = e_{\text{ET}}$ and $4x - p = f_{\text{ET}}$ [8, (2.1), (2.6)]. By [8, (2.7)] the chart quantity is $e = (f_{\text{ET}}e_{\text{ET}} - 1)/4 = a_{\text{ET}}^2 d_{\text{ET}}$. The chart is symmetric in the retained denominators x and pm : a move retaining x keeps a , and a move retaining pm keeps h .

2.5. The seed, the hubs and the bridge.

Lemma 2.8 (PROVED).

- (1) *The denominator $-2pt$ occurs only in σ . Hence every neighbour of σ contains t , and no vertex containing t , nor any positive denominator $z \leq t$, is positive.*
- (2) *The vertices containing t are $V_1(h) = (t, p(ph - t), -t(ph - t)/h)$ for signed $h \mid t^2$, together with the Type II vertices $(t, p(\delta - t), p(t^2/\delta - t))$ for signed $\delta \mid t^2$, $\delta \neq t$. The value $\delta = -t$ gives σ . The third entry of $V_1(h)$ is negative for both signs of h .*
- (3) *The path*

$$\sigma, \quad (t, -t(p + 1), -pt(p + 1)), \quad (2t, 2t + 1, -pt(p + 1)), \quad (2t, 2t, -pt)$$

lies in G_p . The vertices containing $-pt$ are $(-pt, t + d, t + t^2/d)$ for signed $d \mid t^2$, $d \neq -t$.

Proof. (1) A vertex containing $-2pt$ is Type II (its label is $-8t - 1 \equiv 1$), and its other p -divisible denominator pn has $4n - 1 \equiv 1$, so $n \equiv -2t \pmod{p}$. So $|m|, |n| \geq 2t$, and $p/x = 4 - 1/m - 1/n$ gives $x > 0$ and $|4 - p/x| \leq 1/t$. If $x \geq t + 1$ then $4 - p/x \geq 3/(t + 1) > 1/t$. If $x \leq t$ then $p/x - 4 \geq 1/t$, with

equality only for $x = t$, and then $1/m + 1/n = -1/t$ forces $m = n = -2t$. A positive vertex has all denominators $> p/4$.

(2) At $z = t$, $4/p - 1/t = -1/(pt)$. For Type I, the chart with $a = 0$ gives $e = -h$. For Type II, $1/m + 1/n = -1/t$ is $(m+t)(n+t) = t^2$. If $h > 0$ then $ph - t > 0$, and if $h < 0$ then $ph - t < 0$; either way $-t(ph - t)/h < 0$.

(3) The reciprocal sums are checked directly, using $2t + 1 = (p + 1)/2$. Consecutive vertices share t , $-pt(p + 1)$ and $2t$. Finally, $4/p + 1/(pt) = 1/t$, and Lemma 2.1 applies with $r = 1$, $s = t$. $\square$

So the seed component contains the whole fibres of t and of $-pt$. These fibres have $3\tau(t^2)$ and $\tau(t^2)$ vertices. They are the two terms of the unique signed two-term identity $4/p = 1/t - 1/(pt)$.

2.6. Rigid fibres.

Lemma 2.9 (Type II fibres; PROVED). *A Type II denominator pm lies in at most two vertices. If $|m| \geq 2t$, then in every vertex (x, pm, pn) containing it, n lies in $[-2t, 2t]$.*

Proof. Write the vertex as (x, pm, pn) , and put $K = 4m - 1$ and $A = 4 - 1/m \in [3, 5]$. Then $x = p/(A - 1/n)$ and $nK \equiv m \pmod{p}$. The class of n has exactly one representative $n_0 \in [-2t, 2t]$. Every other representative has $|n| \geq (p + 1)/2$, and then, with $\eta = 2/(p + 1)$,

$$\left| x - \frac{pm}{K} \right| = \frac{p/|n|}{A(A - 1/n)} \leq \frac{p\eta}{3(3 - \eta)} = \frac{2p}{9p + 3} < \frac{1}{4}.$$

So all such vertices have the same x , the integer nearest to pm/K . But x determines n , so there are at most two vertices. If $|m|, |n| \geq 2t$, the argument of Lemma 2.8(1) shows that the vertex is σ , where $n = -2t$. $\square$

Theorem 2.10 (outer anchors; PROVED). *Every p -free denominator z with $z < 0$ or $z > 2t$ lies in at most one vertex.*

Proof. By Lemma 2.4 a vertex containing z is of Type I, (z, x, pm) , with x p -free. Each of x and m determines the vertex. As in Lemma 2.4, $m \leq -t$ or $m \geq 3t + 1$. Put $R = 4/p - 1/z$, $\delta_1 = 1/(pt)$ and $\delta_2 = 1/(p(3t + 1))$. Then $1/x = R - 1/(pm) \in [R - \delta_2, R + \delta_1]$.

Case $z < 0$. Here $R > 4/p > \delta_2$ and $R + \delta_1 > 1/t$. So x lies in an interval of length $(\delta_1 + \delta_2)/((R - \delta_2)(R + \delta_1)) < \frac{1}{t(3t+1)} \cdot \frac{p(3t+1)}{12t+3} \cdot t = \frac{1}{3}$.

Case $z \geq p$. Here $R \geq 3/p$. The length is at most $p^2/((3t + 1)(9t + 2)) < 1$.

Case $2t < z < p$. Use the chart of Lemma 2.7 anchored at z . Put $A = 4z - p \in [4t + 3, 3p - 4]$ and $f = z^2/e$, a signed divisor of z^2 . Then $x = Hz^2/e - z$, and $AH = 4e + 1$ gives

$$x + z = Hf, \quad A(x + z) = 4z^2 + f. \quad (2)$$

Every vertex at z satisfies

$$x + z \leq 6t. \quad (3)$$

Indeed, x is the p -free denominator in $[1, 2t]$ given by Lemma 2.4 (since $z \notin [1, 2t]$), and $z \leq p - 1 = 4t$. Equivalently, (2) and $f = (x + z)/H$ give $(x + z)(A - 1/H) = 4z^2$. So $x + z \leq g(z) := 4z^2/(4z - p - 1/3)$, both for $H \geq 3$ and for $H \leq -1$. The function g is convex on $(2t, p)$, so it is bounded by its values at the endpoints. At $z = 2t + 1$ the bound $g \leq 6t$ reads $16t^2 + 16t + 4 \leq 24t^2 + 16t$, and at $z = p - 1$ it reduces to $p \geq 5$. The vertex is positive if and only if $H \geq 3$; then $0 < f \leq (x + z)/3 \leq 2t$ by (3). Otherwise $H \leq -1$ and $f < 0$. By (2), x determines the vertex. Suppose that two vertices share z , with $x_1 < x_2$. Then $f_2 - f_1 = A(x_2 - x_1) \geq A$.

- Both positive: $0 < f_1, f_2 \leq 2t < A$, which is impossible.
- Mixed, positive P and nonpositive N : then $f_P > 0 > f_N$, so $x_P > x_N$. Also $|f_N| \leq x_N + z \leq 6t$ by (3), so $A(x_P - x_N) \leq 8t < 2A$. Hence $x_P = x_N + 1$ and $|f_N| = A - f_P \geq 2t + 3$. If $H_N \leq -5$ then $|f_N| \leq 6t/5$, which is too small. So $H_N = -1$, i.e. $m_N = -t$, and $1/x_N + 1/z = 1/t$. Write $x_N = t + d$ and $z = t + u$ with $du = t^2$ and $d < t < u$. Then $d = \lambda\alpha^2$, $u = \lambda\beta^2$ and $t = \lambda\alpha\beta$ with $\alpha < \beta$; put $\gamma = \beta - \alpha$. A computation from (2) gives

$$\begin{aligned} f_P &= f_N + A = \lambda\gamma(4\beta - \gamma) - 1, \\ x_P + z &= w + 1, \quad z^2 = \lambda\beta^2 w, \quad w := \lambda(\alpha + \beta)^2. \end{aligned}$$

Positivity needs $f_P \mid z^2$ and $f_P \mid x_P + z = w + 1$. So $\gcd(f_P, w) = 1$, and then $f_P \mid \lambda\beta^2$. As $f_P \equiv -1 \pmod{\lambda}$, in fact $f_P \mid \beta^2$, say $\beta^2 = kf_P$ with $k \geq 1$. Then β is a root of $X^2 - 4k\lambda\gamma X + k(\lambda\gamma^2 + 1)$. So the quarter-discriminant $k(\lambda\gamma^2\mathcal{M} - 1)$, with $\mathcal{M} = 4k\lambda - 1 \geq 3$, is a square s^2 , and $s^2 \equiv -k \pmod{\mathcal{M}}$ with $\gcd(k, \mathcal{M}) = 1$. But $(-1|\mathcal{M}) = -1$. If $2 \mid k$ then $\mathcal{M} \equiv 7 \pmod{8}$, so $(2|\mathcal{M}) = 1$. For the odd part k' of k , reciprocity and $\mathcal{M} \equiv -1 \pmod{k'}$ give $(k'|\mathcal{M}) = (-1|k')(-1)^{(k'-1)/2} = 1$. So the Jacobi symbol $(-k|\mathcal{M}) = -1$, and $-k$ is not a square modulo $\mathcal{M}$; this is a contradiction.

- Both nonpositive: $|f_1| = |f_2| + A(x_2 - x_1) \geq 4t + 4$, so, as above (using (3)), $H_1 = -1$, $x_1 = t + d$, $z = t + u$. Then $|f_2| \leq x_1 + z - A = 2t + d - 3u + 1 < 1$, which is impossible. $\square$

The cases $z < 0$ and $z \geq p$, and a bound of two vertices for $2t < z < p$, were proved first, by interval estimates, in the signed-refactor notes. These predate the separation of this project from the companion project [1], and both projects contain them. The Vieta–Jacobi step closing the case $2t < z < p$ was found later, in this project only. Theorem 2.10 was brute-force checked for all $p \equiv 1 \pmod{4}$, $p < 3 \cdot 10^4$, and independently for $p < 400$ on all $z \in [-4000, -1] \cup [2t + 1, 4000]$; no violation was found.

Corollary 2.11 (where sign changes happen; PROVED). *If a positive and a nonpositive vertex share a denominator z , then either z is p -free with $t < z \leq 2t$, or $z = p(ph - t)$ is a Type I denominator with $h \geq 1$.*

Proof. z lies in a positive vertex, so $z > p/4 > t$. If z is p -free, then $z \leq 2t$ by Theorem 2.10. If z is p -divisible, then it is of Type I by Corollary 2.6, and $h \geq 1$ because $z > 0$. $\square$

3. SHORT ESCAPES, FORCED EXITS AND SIEVE BOUNDS

A positive p -free denominator z is *productive* if its fibre contains a positive vertex. For $t < z \leq 2t$ put $q_z = 4z - p > 0$. By Lemma 2.1, applied with $r = q_z$ and $s = pz$, the denominator z is productive if and only if some positive $D \mid z^2$ satisfies one of

$$D \equiv -z \pmod{q_z} \quad (\text{Type II exit } (z, p(z+D)/q_z, p(z+z^2/D)/q_z)), \quad (4)$$

$$D \equiv -1/4 \pmod{q_z} \quad (\text{Type I exit } (z, p(z+pD)/q_z, (pz+z^2/D)/q_z)). \quad (5)$$

This is the classical anchor criterion: (5) is the Type I condition $p = 4acd - f$, $f \mid 4a^2d + 1$ of [8, Prop. 2.3], and (4) is the corresponding Type II condition of [8, Prop. 2.7], written in the anchor z .

Theorem 3.1 (escapes of length ≤ 3 ; PROVED). *Let $p = 4t + 1$ be prime.*

(1) $\text{dist}(p) \geq 2$, and $\text{dist}(p) \leq 2$ if and only if

$$\exists h > 0, h \mid t^2, \exists D > 0, D \mid (ph - t)^2 : D \equiv -h \pmod{4h - 1}. \quad (6)$$

(2) $\text{dist}(p) \leq 3$ if and only if (6), (A) or (B) holds:

- (A) (negative-bucket transfer) *there are $c > 0$ with $c \mid t^2$, and $d > 0$ with $d \mid (pc + t)^2$ and $d \equiv -c \pmod{4c + 1}$, such that, with $a = (d + c)/(4c + 1)$, $x = t + a$ and $w = x(pc + t)/d$, one of x, w is productive. The path is $\sigma \rightarrow (t, -p(pc + t), -t(pc + t)/c) \rightarrow (x, -p(pc + t), w) \rightarrow$ positive;*
- (B) (Type II swap) *for some signed $\delta \mid t^2$, $\delta \neq \pm t$, and some $m \in \{\delta - t, t^2/\delta - t\}$, the other vertex (x', pm, pn') of the bucket of pm (Lemma 2.9) exists and its p -free denominator x' is productive.*

Proof. The first layer L_1 consists of the vertices containing t (Lemma 2.8(1)), and all of them are nonpositive.

Distance two. Let P be a positive neighbour of $V \in L_1$. It shares some $z \neq t$ with V , by Lemma 2.8(1). If $V = V_1(h)$, then z is not the negative entry, so $z = p(ph - t)$ with $h > 0$, and P is a positive vertex of chart column h . By Lemma 2.7 this means that some $a \geq 1$ has $e = (4a - 1)h - a$ dividing m^2 , $m = ph - t$. Here $e \equiv -h \pmod{4h - 1}$, and conversely every positive $D \equiv -h$ is of the form e with $a = (D + h)/(4h - 1) \geq 1$. This is (6). If V is of Type II, then it is nonpositive. So both of its labels are residues by Corollary 2.6, and both of its p -divisible buckets are nonpositive.

Distance three. Let σ, V_1, V_2, P be a shortest path with $V_1 \in L_1$, V_2 nonpositive and not containing t . Let $z_1 \neq t$ be shared by V_1 and V_2 . By Lemma 2.3, V_2 has the type of the bucket of z_1 .

- $V_1 = V_1(h)$ and z_1 its negative entry: then $V_2 = V_1$ by Theorem 2.10, which is impossible.
- $V_1 = V_1(h)$, $h > 0$, $z_1 = p(ph - t)$: chart V_2 in column h from its p -free denominator $x_2 \in [1, 2t]$ (Lemma 2.4). Nonpositivity forces $a \leq 0$, so $x_2 \leq t$ and $e = a(4h - 1) - h < 0$, and the third entry is negative. So P shares $p(ph - t)$ with V_2 , and then P is adjacent to V_1 , contradicting minimality.
- $V_1 = V_1(-c)$, $c > 0$, $z_1 = -p(pc + t)$: chart V_2 in column $h = -c$. If $a \leq 0$, then $e = c(1 - 4a) - a > 0$, the third entry is negative and $x_2 \leq t$, so V_2 has no positive neighbour. If $a \geq 1$ then $d := -e = (4c + 1)a - c > 0$, and both p -free entries are positive. By Lemma 2.7(2), $d \mid (t + a)^2$ if and only if $d \mid (pc + t)^2$. This is (A).
- V_1 of Type II and $z_1 = pm$: then V_2 is the other vertex of that bucket, and it is nonpositive (Corollary 2.6). The label of its second p -divisible denominator is a residue, so that bucket is nonpositive too. Hence P contains x' . This is (B).

Conversely, every path listed in (A) and (B) is legal. $\square$

The proof was checked against brute-force layered BFS: for all $p \equiv 1 \pmod{4}$ with $p < 3 \cdot 10^4$, for every fifth such prime up to $3 \cdot 10^5$, and independently for all $p < 2500$. The sets of positive vertices reached by escapes of length exactly two and exactly three coincide with those predicted by the theorem.

Lemma 3.2 (exits need a fresh non-residue; PROVED).

- (1) A positive exit of (6) with certificate D requires $\left(\frac{D}{p}\right) = -1$. Hence $ph - t$ has a prime factor ℓ with $\left(\frac{\ell}{p}\right) = -1$.
- (2) At an anchor $z > t$ with certificate $D \mid z^2$, the Type I exit (5) requires $\left(\frac{D}{p}\right) = -1$, and the Type II exit (4) requires $\left(\frac{zD}{p}\right) = -1$.

Every prime $\ell \mid t$ has $\left(\frac{\ell}{p}\right) = 1$. So the required non-residue comes from a prime factor of a new shift $(ph - t, pc + t, t + a, \dots)$, not from the support of t .

Proof. (1) With $K = 4h - 1$ and $m = ph - t$, the p -free pair is $y = (m + D)/K$, $w = (m + m^2/D)/K$, and $yw = m(m + D)^2/(DK^2)$. Since $m \equiv 1/4 \pmod{p}$, Theorem 2.5 gives the claim. (2) The p -free pair of (5) has product $z^2(z + pD)/(Dq_z)$, and $q_z \equiv 4z \pmod{p}$. For (4) the quotient pair has product $z(z + D)^2/(Dq_z^2)$. Finally, if $\ell \mid t$ then $p \equiv 1 \pmod{\ell}$, so $\left(\frac{\ell}{p}\right) = 1$ by reciprocity (and $p \equiv 1 \pmod{8}$ if $2 \mid t$). $\square$

Lemma 3.3 (forced exits; PROVED).

- (1) If $p \equiv 5 \pmod{8}$, then (6) holds with $h = 1$, $D = 2$.
- (2) If $p \equiv 17 \pmod{24}$, then (6) holds with $h = 2$, $D = 12$.

- (3) If $p \equiv 1 \pmod{24}$ and p is a quadratic non-residue modulo 5 or modulo 7, then (6) holds as follows ($m = ph - t$):

class	h	D	reason
$p \equiv 3 \pmod{5}$	1	5	$t \equiv 3 \pmod{5}$, $5 \mid 3t + 1$, $5 \equiv -1 \pmod{3}$
$p \equiv 2 \pmod{5}$	2	5	$t \equiv 4 \pmod{5}$, $5 \mid 7t + 2$, $5 \equiv -2 \pmod{7}$
$p \equiv 3 \pmod{7}$	6	63	$m = 23t + 6$, $21 \mid m$, $63 \equiv -6 \pmod{23}$
$p \equiv 5 \pmod{7}$	3	63	$m = 11t + 3$, $21 \mid m$, $63 \equiv -3 \pmod{11}$
$p \equiv 6 \pmod{7}$	4	$m/14$	$m = 15t + 4$, $14 \mid m$, $m/14 \equiv -4 \pmod{15}$

- (4) (X-branch.) Let $d \mid t^2$, $d > 0$, and $z = t + d$. If some positive $D \mid z^2$ has $D \equiv -1/4 \pmod{4d - 1}$, then $\text{dist}(p) \leq 5$, via the bridge of Lemma 2.8(3), the vertex $(-pt, z, tz/d)$ and the Type I exit (5) at z .

Proof. (1) t is odd, $m = 3t + 1$ is even, and $2 \equiv -1 \pmod{3}$. (2) t is even and $t \equiv 1 \pmod{3}$, so $6 \mid 7t + 2 = 2p - t$ and $12 \mid m^2$, with $12 \equiv -2 \pmod{7}$. (3) In each row $6 \mid t$, so $h \mid t^2$. The stated congruences are immediate (for the last row, $14 \equiv -1 \pmod{15}$). (4) z is p -free by Lemma 2.3, since $-pt$ has label 0. Its modulus is $q_z = 4d - 1$. $\square$

Corollary 3.4 (Mordell's classes; PROVED). *Every prime $p \equiv 1 \pmod{4}$ outside the six classes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$ has $\text{dist}(p) = 2$.*

These are the classical hard classes [13], so the corollary adds nothing to ES. It shows that the seed reproduces Mordell's identities in two moves, and that every question about the seed concerns only these classes.

Lemma 3.5 (half-dimension; adapted from [5, Lemma 4.2]). *Let $K \geq 3$, $c \in (\mathbb{Z}/K)^\times$, and let $N \geq 1$ be coprime to K . If no positive divisor of N^2 is $\equiv c \pmod{K}$, then the set R of residues modulo K of the prime factors of N satisfies $R \cap cR^{-1} = \emptyset$. In particular $|R| \leq \varphi(K)/2$.*

Proof. If $r, r' \in R$ with $rr' \equiv c$, choose primes $\ell \equiv r$ and $\ell' \equiv r'$ dividing N . If $\ell \neq \ell'$ then $\ell\ell' \mid N$; otherwise $\ell^2 \mid N^2$. Either way some divisor of N^2 lies in the class c . $\square$

Theorem 3.6 (PROVED, given a standard upper-bound sieve). *For primes $p \equiv 1 \pmod{4}$,*

$$\#\{p \leq N : \text{dist}(p) > 2\} \ll N(\log N)^{-11/2},$$

$$\#\{p \leq N : \text{dist}(p) > 5\} \ll N(\log N)^{-10}.$$

Proof. By Lemma 3.3, $\text{dist}(p) > 2$ forces p into one of the six classes modulo 840. Then $6 \mid t$, so every $h \mid 36$ and every $d \mid 36$ divides t^2 . Consider the nine branches $N_h = ph - t$, $K_h = 4h - 1$, $c_h = -h$ ($h \mid 36$), and the nine branches $N_d = t + d$, $K_d = 4d - 1$, $c_d = -1/4$ ($d \mid 36$). We have $\gcd(N_h, K_h) = 1$, and $\gcd(N_d, K_d) = 1$ for $p > K_d$, because $4N_h = pK_h + 1$ and $4N_d = p + K_d$. If $\text{dist}(p) > 2$, then each h -branch fails. If $\text{dist}(p) > 5$, then each d -branch also

fails (Lemma 3.3(4)). By Lemma 3.5, a failing branch b confines the residues of the prime factors of N_b to a set R_b with $|R_b| \leq \varphi(K_b)/2$.

Fix one admissible family $(R_b)_b$; there are finitely many. For a prime ℓ outside a finite exceptional set, $\ell \mid N_h$ if and only if $p \equiv -K_h^{-1} \pmod{\ell}$, and $\ell \mid N_d$ if and only if $p \equiv -K_d \pmod{\ell}$. For large ℓ these classes are pairwise distinct and different from 0, since $(1 - 4d)(4h - 1) + 1 = 4(h + d - 4dh) \neq 0$. Let $\mathcal{L}$ be the set of primes $\ell > 19$ that do not divide $840 \prod_b K_b$ and are not coincidence primes. Discard the primes $p \leq N^{1/2}$, which are negligible. The remaining primes counted lie in $\mathcal{A} = \{n \leq N : n \equiv a \pmod{840}\}$ for a fixed a . For each $\ell \in \mathcal{L}$ with $\ell \leq N^{1/2}$, they avoid the class 0 and the class of every branch b with $\ell \bmod K_b \notin R_b$; these classes are distinct. So they survive the sieve of $\mathcal{A}$ by $\mathcal{L}$, up to $z = N^{1/2}$, with

$$\omega(\ell) = 1 + \#\{b : \ell \bmod K_b \notin R_b\} \quad (\ell \in \mathcal{L}).$$

Here $\omega(\ell) \leq 1 + 18 < \ell$. For squarefree d composed of primes in $\mathcal{L}$, $|\mathcal{A}_d| = \omega(d)X/d + r_d$ with $X = N/840$ and $|r_d| \leq \omega(d)$. By the prime number theorem in progressions modulo each K_b ,

$$\sum_{\substack{w \leq \ell < z \\ \ell \in \mathcal{L}}} \frac{\omega(\ell) \log \ell}{\ell} = \kappa \log \frac{z}{w} + O(1), \quad \kappa = 1 + \sum_b \left(1 - \frac{|R_b|}{\varphi(K_b)}\right) \geq 1 + \frac{B}{2},$$

where $B = 9$, respectively 18, is the number of branches. These are the hypotheses (Ω_1) , $(\Omega_2(\kappa))$ and (R) of the upper-bound sieve of dimension κ (e.g. [9, Thm. 2.2]), which gives $\ll X \prod_{\ell < z, \ell \in \mathcal{L}} (1 - \omega(\ell)/\ell) \ll N(\log N)^{-\kappa}$. Summing over the finitely many choices of (R_b) proves the theorem. $\square$

The case $h = 1$ alone gives the exponent $3/2$. It is the direct analogue of Dahan's upper bound [5, Thm. 4.3] with a single pair; Dahan's Theorem 4.14 is a two-sided estimate $\asymp N(\log N)^{-3/2}$ for a different obstruction (shift $c = 7$). These savings are only polylogarithmic. Theorem 4.9 shows that, under Bateman–Horn, no bounded depth can do better than polylogarithmic. The data (§6) suggest that the exponents are not sharp.

4. FORMAL GENERICITY

4.1. Hypotheses. Let $\mathcal{P}$ be the set of irreducible $g \in \mathbb{Z}[X]$ with content 1 and positive leading coefficient, and put $P(X) = 24X + 1$.

Hypothesis 4.1 (H; Schinzel–Sierpiński [15]). *If $f_1, \dots, f_m \in \mathcal{P}$ and no prime divides $f_1(y) \cdots f_m(y)$ for all $y \in \mathbb{Z}$, then there are infinitely many y with all $f_i(y)$ prime.*

Dickson's conjecture [6] is the case where all f_i are linear. The *Bateman–Horn conjecture* [3] predicts that the number of such $y \leq Y$ is asymptotic to $C Y (\log Y)^{-m} / \prod_i \deg f_i$ for a constant $C > 0$. Each result below uses only a finite, explicit instance.

4.2. Formal frames.

Definition 4.2. A *frame* $\mathfrak{F} = (\Lambda, E, q_0, S_e, S_a)$ consists of a finite set $\Lambda \supseteq \{2, 3\}$ of primes, exponents $E_\ell \geq 1$ ($\ell \in \Lambda$), an integer q_0 , and finite disjoint sets $S_e, S_a \subset \mathcal{P}$ (*entry* and *auxiliary* polynomials) with $X, P \in S_e$. For $g \in S = S_e \cup S_a$ we require $g(q_0) \neq 0$ and $E_\ell > v_\ell(g(q_0))$, and we put

$$C_g = \prod_{\ell \in \Lambda} \ell^{v_\ell(g(q_0))}, \quad M = \prod_{\ell \in \Lambda} \ell^{E_\ell}.$$

An integer q is *admissible* if $q \equiv q_0 \pmod{M}$ and the numbers $r_g = g(q)/C_g$, $g \in S_e$, are pairwise distinct primes outside Λ . A *formal integer* is $\varphi = c \prod_{g \in S_e} (g/C_g)^{e_g}$ with $c \in \mathbb{Z} \setminus \{0\}$ and $e_g \geq 0$. Its value at q is $\varphi(q)$, and its *sign* is the sign of c . A *formal vertex* is a triple of formal integers with $1/a + 1/b + 1/c = 4/P$ in $\mathbb{Q}(X)$. It is *formally positive* if all three constants are positive.

Lemma 4.3 (PROVED). *Let $q \equiv q_0 \pmod{M}$ and $g \in S$. Then $n_g(q) := g(q)/C_g$ is an integer prime to every $\ell \in \Lambda$, and $n_g(q) \rightarrow \infty$ as $q \rightarrow \infty$. Now suppose that $C_X = C_P = 1$, fix a finite set Φ of formal integers, and let q be admissible and larger than a threshold depending on Φ . Then:*

- (1) q and $p = 24q + 1$ are prime;
- (2) the value of each $\varphi \in \Phi$ is a nonzero integer with the sign of its constant, and its p -adic valuation is the exponent of P ;
- (3) distinct members of Φ have distinct values;
- (4) the values of a formal vertex with entries in Φ form a vertex of G_p .

Proof. $v_\ell(g(q) - g(q_0)) \geq E_\ell > v_\ell(g(q_0))$, so $v_\ell(g(q)) = v_\ell(C_g)$. At admissible q the value is $c \prod r_g^{e_g}$, a factorisation into the distinct primes r_g , and $r_X = q$, $r_P = p$. Choose the threshold so that every r_g exceeds the absolute values of all constants of members of Φ . Then the factorisations of the values are read off from the formal expressions, which gives (2) and (3). Signs: every $g \in \mathcal{P}$ has positive leading coefficient. $\square$

Throughout this section, “for large admissible q ” means: for admissible q beyond a threshold depending on the finitely many formal expressions involved (constants, collision and sign bounds, and the resultant bounds in the proof of Lemma 4.5). Every application below involves finitely many expressions.

Lemma 4.4 (H gives admissible values; CONDITIONAL). *Assume that $C_X = C_P = 1$ and*

- (E3) *for every prime $\ell \notin \Lambda$ with $\ell \leq \sum_{g \in S_e} \deg g$, some residue class modulo ℓ is a root of no $g \in S_e$.*

Put $f_g(y) = g(My + q_0)/C_g$ for $g \in S_e$. Then the f_g lie in $\mathcal{P}$ and have no common fixed prime divisor. Under Hypothesis H for $\{f_g\}$ there are infinitely many admissible q . Under Bateman–Horn there are $\gg N(\log N)^{-|S_e|}$ admissible q with $24q + 1 \leq N$.

Proof. The Taylor coefficients of $g(My + q_0)$ of positive degree are divisible by M , and $C_g \mid M$, while $C_g \nmid g(q_0)$; so $f_g \in \mathbb{Z}[y]$. It is irreducible with positive leading coefficient. If $\ell \in \Lambda$, then $f_g(y) \equiv f_g(0) \not\equiv 0 \pmod{\ell}$, since $v_\ell(g(My + q_0) - g(q_0)) \geq E_\ell > v_\ell(C_g)$. This shows that f_g is primitive at ℓ and has no root modulo ℓ . If $\ell \notin \Lambda$, then $y \mapsto My + q_0$ is invertible modulo ℓ and $\ell \nmid C_g$. So f_g is primitive at ℓ , and by (E3), or by degree counting when $\ell > \sum \deg g$, $\prod f_g$ has a non-root modulo ℓ . For large y the prime values are pairwise distinct, because distinct elements of $\mathcal{P}$ are not proportional. They lie outside Λ , because they are ℓ -units. $\square$

4.3. Formal fibres. Let Z be a formal integer. Then $N := 4Z - P \neq 0$. Cancel common irreducible factors in $N/(PZ)$ and write the result as r/s with (here h runs over the irreducible factors of N that are not cancelled; C_h is defined once $h \in S$, see (F1) below)

$$r = c_r \prod_h (h/C_h)^{\beta_h}, \quad s = c_s \prod_{g \in S_e} (g/C_g)^{\alpha_g}, \quad \gcd(c_r, c_s) = 1, \quad c_s > 0.$$

The polynomials of s are those of PZ , hence lie in S_e . A *formal divisor* of s^2 is $D = \pm d \prod (g/C_g)^{j_g}$ with $d \mid c_s^2$ and $0 \leq j_g \leq 2\alpha_g$. For such D consider

- (A) $h^{\beta_h} \mid D + s$ in $\mathbb{Q}[X]$ for every h occurring in r ;
- (B) writing $D + s = k \prod_h h^{\beta_h}$ and $K = k \prod_h C_h^{\beta_h}$, $v_\ell(K(q_0)) \geq v_\ell(c_r)$ for every $\ell \mid c_r$.

Let $\text{Fib}(Z)$ be the set of $D \neq -s$ such that both D and s^2/D satisfy (A) and (B). For $D \in \text{Fib}(Z)$ put $y_D = (D + s)/r$.

Lemma 4.5 (formal fibres; PROVED). *Suppose that $C_X = C_P = 1$ and*

- (F1) *every irreducible factor of N lies in S ;*
- (F2) *every prime of c_r lies in Λ , and $E_\ell \geq v_\ell(c_r) + v_\ell(\text{den}(D + s))$ for all formal divisors D and all $\ell \mid c_r$.*

Then for all large admissible q , the vertices of G_p containing $Z(q)$ are exactly the triples $(Z(q), y_D(q), y_{s^2/D}(q))$ with $D \in \text{Fib}(Z)$.

Proof. By Lemma 2.1, applied to the possibly non-reduced fraction $r(q)/s(q)$ (the identity $(ry - s)(rw - s) = s^2$ does not need reducedness), every vertex containing $z = Z(q)$ is $(z, (s + D)/r, (s + s^2/D)/r)$ at q for a divisor D of $s(q)^2$. By Lemma 4.3, $s(q) = c_s \prod r_g^{\alpha_g}$ with distinct primes $r_g > |c_s|$. So its signed divisors are exactly the values of the formal divisors, and $D = -s$ is excluded. It remains to show that, for a fixed formal divisor $D \neq -s$ and large admissible q , the number $y_D(q)$ is an integer exactly when (A) and (B) hold.

(A) *is necessary.* Suppose $D + s = h^j k'$ with $j < \beta_h$ and $h \nmid k'$. Put $n = n_h(q)$, which is prime to Λ by Lemma 4.3. If $y_D(q) \in \mathbb{Z}$, then n^{β_h} divides the integer $(D + s)(q) = y_D(q) r(q) = C_h^j n^j k'(q)$, because $r(q) = c_r \prod n_{h'}(q)^{\beta_{h'}}$. Let d' be the denominator of k' . By Gauss's lemma it equals the denominator of $D + s$, so it is Λ -supported. Then $d'k' \in \mathbb{Z}[X]$, and since $\pi \nmid d'C_h$ for

every prime $\pi \mid n$, we get $v_\pi(d'k'(q)) \geq (\beta_h - j)v_\pi(n) \geq v_\pi(n)$. Also $v_\pi(h(q)) \geq v_\pi(n)$. Writing $uh + vd'k' = \text{Res}(h, d'k') \neq 0$ with $u, v \in \mathbb{Z}[X]$, we get $n \mid \text{Res}$, which fails once n is large.

Given (A), (B) is exact. Here $y_D(q) = K(q) \prod n_h^{\beta_h} / (c_r \prod n_h^{\beta_h}) = K(q)/c_r$. The number $K(q)$ is an integer: its denominator is Λ -supported, and $K(q) \prod n_h^{\beta_h} = (D + s)(q) \in \mathbb{Z}$ with n_h prime to Λ . So $y_D(q) \in \mathbb{Z}$ if and only if $v_\ell(K(q)) \geq v_\ell(c_r)$ for all $\ell \mid c_r$. By (F2), $K(q) \equiv K(q_0)$ to the precision needed.

The same applies to s^2/D . Finally, $y_D \neq 0$ for large q , because $D + s \neq 0$. $\square$

A formal integer Z is *dead* if it is p -free (its P -exponent is 0) and either its constant is negative or $Z - 12X$ has positive leading coefficient. By Lemma 4.3, at large admissible q its value is a p -free denominator outside $[1, 2t]$. By Theorem 2.10 it then lies in at most one vertex, so dead denominators never need to be expanded.

4.4. Formal positivity is impossible in a square class.

Proposition 4.6 (Schinzel; Bright–Loughran; PROVED). *Let $\mathfrak{F}$ be a frame such that $24q_0 + 1$ is a nonzero square modulo every $\ell \in \Lambda \setminus \{2, 3\}$. Assume that Λ contains the primes of $H_g := 24^{\deg g}g(-1/24) \in \mathbb{Z} \setminus \{0\}$ for every $g \in S_e \setminus \{P\}$. Let V be a formal vertex whose three constants are Λ -supported up to sign. Then V is not formally positive.*

Proof. The class $24q_0 + 1 \pmod{24M}$ is a unit, so by Dirichlet's theorem there are arbitrarily large $q \equiv q_0 \pmod{M}$ with $p = 24q + 1$ prime. By Lemma 4.3, the values $n_g(q)$ are integers, so $V(q)$ is a vertex of G_p (no primality of $n_g(q)$ is needed here). Since $24q \equiv -1 \pmod{p}$, we have $24^{\deg g}g(q) \equiv H_g \pmod{p}$. So $p \nmid n_g(q)$ for $g \neq P$ once $p > |H_g|$, and the p -adic valuation of each entry is its P -exponent. The equal-valuation pair, with p removed, is therefore $\pm$ a product of primes of Λ and of numbers $n_g(q)$, $g \neq P$. For $\ell \in \Lambda$, $\ell \geq 5$, we have $\left(\frac{\ell}{p}\right) = \left(\frac{p}{\ell}\right) = \left(\frac{24q_0+1}{\ell}\right) = 1$, while $\left(\frac{2}{p}\right) = \left(\frac{3}{p}\right) = \left(\frac{-1}{p}\right) = 1$ because $p \equiv 1 \pmod{24}$. Also $\left(\frac{n_g(q)}{p}\right) = \left(\frac{g(q)}{p}\right) = \left(\frac{H_g}{p}\right) = 1$, because 24 , C_g and H_g are products of such primes. By Theorem 2.5, $V(q)$ is nonpositive. At large q the signs of the entries are the signs of the constants, so some constant is negative. $\square$

A formal vertex is a polynomial solution of (1) in $n = 24X + 1$, integral on the class $n \equiv 24q_0 + 1 \pmod{24M}$, and that class is a square class. Proposition 4.6 is thus a form of Schinzel's theorem that no square class is solvable by polynomials with positive leading coefficients [14], [8, §1]; the proof is the standard reciprocity argument, here through Theorem 2.5.

4.5. Closed formal explorations.

Theorem 4.7 (closed explorations are sterile; CONDITIONAL). *Let $\mathfrak{F}$ be a frame and $\mathcal{V}$ a finite set of formal vertices such that*

- (E1) $C_X = C_P = 1$ and $(6X, -12XP, -12XP) \in \mathcal{V}$;
- (E2) *for every entry Z of every member of $\mathcal{V}$ that is not dead, (F1) and (F2) hold, and for every $D \in \text{Fib}(Z)$ the functions $y_D, y_{s^2/D}$ are formal integers with $(Z, y_D, y_{s^2/D}) \in \mathcal{V}$;*
- (E3) *as in Lemma 4.4;*
- (E4) *no member of $\mathcal{V}$ is formally positive.*

Then for every large admissible q the seed component of $p = 24q + 1$ lies in the value set of $\mathcal{V}$ and has no positive vertex. If $\mathcal{V}$ is connected under shared formal entries, then the seed component equals this value set. Under Hypothesis H for $\{f_g : g \in S_e\}$ there are infinitely many such p , and under Bateman–Horn $\gg N(\log N)^{-|S_e|}$ of them up to N .

Proof. The seed is the value of $(6X, -12XP, -12XP)$. Let $v = V(q)$ with $V \in \mathcal{V}$, and let v' be a neighbour sharing $z = Z(q)$, where Z is an entry of V . If Z is dead, its fibre is $\{v\}$. Otherwise Lemma 4.5 and (E2) show that v' is a value of a member of $\mathcal{V}$. By induction on the distance, the seed component lies in the value set. By (E4) and Lemma 4.3, every value is nonpositive. Formal vertices sharing a formal entry have values sharing a denominator, which gives the equality. The last sentence is Lemma 4.4. $\square$

By Proposition 4.6, (E4) holds automatically in a square class once Λ contains the primes of the constants and of the H_g . In that case the only thing to check is the closure (E2). This is the precise form of the Principle for explorations of G_p : at an admissible q , any exploration whose formal shadow is closed meets only formal vertices, and none of them is positive.

4.6. Bounded radius: unbounded seed distance.

Lemma 4.8 (a generic base point; PROVED). *There is $q^* = (q_\ell^*) \in \prod_\ell \mathbb{Z}_\ell$ such that (1) each q_ℓ^* is a unit and each $24q_\ell^* + 1$ is a nonzero square unit in $\mathbb{Z}_\ell$; (2) $g(q_\ell^*) \neq 0$ for every $g \in \mathcal{P}$ and every ℓ ; (3) for each $g \in \mathcal{P}$, $v_\ell(g(q_\ell^*)) = 0$ for all but finitely many ℓ .*

Proof. For $\ell = 2, 3$ take a transcendental unit; $24u + 1 \equiv 1$ modulo 8, respectively 3, is a square. Enumerate $\mathcal{P} = \{g_1, g_2, \dots\}$. For $\ell \geq 5$ let $n(\ell)$ be maximal with $\sum_{i \leq n(\ell)} \deg g_i < (\ell - 3)/2$. There are $(\ell - 3)/2$ residues $r \not\equiv 0$ with $24r + 1$ a nonzero square. Each primitive g_i has at most $\deg g_i$ roots modulo ℓ , so some such r is a root of none of $g_1, \dots, g_{n(\ell)}$. Lift r to a transcendental element of $r + \ell\mathbb{Z}_\ell$. Since $n(\ell) \rightarrow \infty$, (3) follows. $\square$

For $g \in \mathcal{P}$ put $C_g = \prod_\ell \ell^{v_\ell(g(q_\ell^*))}$. Any frame whose Λ contains the primes of these C_g ($g \in S$), and with $q_0 \equiv q_\ell^* \pmod{\ell^{E_\ell}}$, has the same constants C_g .

Theorem 4.9 (unbounded seed distance; CONDITIONAL on H). *For every $k \geq 0$ there is a finite set $S_e^{(k)} \subset \mathcal{P}$, depending on k and on the base point of Lemma 4.8, with the following property. Under Hypothesis H for the*

associated $\{f_g\}$, there are infinitely many primes q with $p = 24q + 1$ prime such that every vertex within distance k of σ is nonpositive. In particular $\text{dist}(p) > k$. Under Bateman–Horn there are $\gg_k N(\log N)^{-|S_e^{(k)}|}$ such $p \leq N$.

Proof. Fix q^* as in Lemma 4.8. We build frames $\mathfrak{F}_j$ compatible with q^* and sets $\mathcal{V}_j$, by induction. Start with $\mathcal{V}_0 = \{(6X, -12XP, -12XP)\}$, $S_e = \{X, P\}$, $S_a = \emptyset$ and $\Lambda = \{2, 3\}$. At step j , process every entry Z of every member of $\mathcal{V}_j$, and enlarge the data as follows:

- add the factors of $4Z - P$ to S_a (unless they are in S_e), and add the primes of c_r to Λ ;
- compute $\text{Fib}(Z)$. Condition (A) does not depend on q . Condition (B) is decided by q^* , after the E_ℓ have been raised to the precision required by (F2). Add the irreducible factors of every y_D to S_e (removing them from S_a if necessary). Let $\mathcal{V}_{j+1} \supseteq \mathcal{V}_j$ consist of $\mathcal{V}_j$ and all these triples;
- add to Λ the primes of all new C_g , of all constants of entries, of all H_g , and all primes $\leq \sum_{S_e} \deg g$;
- raise the E_ℓ to satisfy Definition 4.2 and (F2), and let q_0 be the CRT lift of q^* .

Each y_D is a formal integer. Its constant c_y is rational. By the proof of Lemma 4.5, the value $y_D(q) = c_y \prod n_g(q)^{e_g}$ is an integer for every large $q \equiv q_0 \pmod{M}$, and no primality is needed for this. A prime in the denominator of c_y cannot lie in Λ , since the $n_g(q)$ are prime to Λ . It cannot lie outside Λ either: such a prime exceeds $\sum \deg g$, so it fails to divide $\prod n_g(q)$ for some such q . Admissibility for $\mathfrak{F}_{j+1}$ implies admissibility for $\mathfrak{F}_j$. By Lemma 4.5, every neighbour of a value of $\mathcal{V}_j$ is a value of $\mathcal{V}_{j+1}$. So after k steps every vertex within distance k of σ is a value of $\mathcal{V}_k$, for large $\mathfrak{F}_k$ -admissible q . The class $24q_0 + 1$ is a square class by Lemma 4.8(1), so Proposition 4.6 gives (E4) for $\mathcal{V}_k$. Condition (E3) is vacuous, because the small primes lie in Λ . Lemma 4.4 finishes the proof. $\square$

Remark 4.10. Only the entry polynomials need to take prime values; the auxiliary factors of $4Z - P$ need not. The proof uses nothing about σ except that its entries are formal. So the same conclusion holds for any bounded-radius exploration from formally given vertices, for instance any “universal” path built from divisors that exist for all p in a residue class. Schinzel’s theorem and [8, Prop. 1.6] rule out arguments that never factor. Theorem 4.9 rules out, under H, bounded-depth searches that do factor. It is the H-conditional form of the Elsholtz–Tao remark [8, §1]. A related unconditional result for one branch is Dahan’s [5, Thm. 4.14]: the shift $c = 7$ fails for $\asymp N(\log N)^{-3/2}$ primes. His computed list [5, Prop. 3.8] of 14 primes below $5 \cdot 10^9$ that fail the modulator-one family is finite evidence of the same kind. The family $S_e^{(k)}$ is large, so the theorem gives no example within computational reach.

4.7. The whole seed component. Theorem 4.9 says nothing about the full component, which is finite for each p but whose radius is not bounded uniformly. Two independent constructions close it. We state the stronger one first. It is *reported* from the companion project, and this paper does not prove it. Within this paper, the statement “Conjecture 1.2 is false under Hypothesis H” rests on Theorem 4.13 alone.

Theorem 4.11 (REPORTED [1]; conditional on Dickson). (Status: the finite packet is re-checked here; the eventual-closure argument is not re-verified here.) *There are explicit integers $Q, M > 0$ and explicit primitive linear forms $F_1, \dots, F_{159}$, $F_i(n) = (\alpha_i(Q + Mn) + \beta_i)/C_i$, with the following property. The forms are $q = Q + Mn$, $p = 24q + 1$, and the primitive parts of $6q + a$ ($a \in A$, $|A| = 84$) and of $6(4h - 1)q + h$ ($h \in H$, $|H| = 75$), which correspond to the shifted anchors $t + a$ and the Type I quotients $ph - t$, with q counted once.*

- (1) *The tuple (F_i) is locally admissible. There is a computable N_0 such that, if $n \geq N_0$ and all $F_i(n)$ are prime, then the seed component of p contains no positive vertex.*
- (2) *On the admissible subprogression $n \equiv 507 \pmod{857}$, where $q \equiv 713 \pmod{857}$,*

$$\left(6q + 7, (24q + 1)(762q + 32), \frac{(6q + 7)(762q + 32)}{857}\right)$$

is a positive solution. It is the Type I chart point $a = 7$, $h = 32$, $e = 857$. Under the hypotheses of (1) it lies outside the seed component.

Consequently, according to [1], under Dickson’s conjecture for these 159 forms there are infinitely many primes with a sterile seed component. Under Dickson’s conjecture for the restricted tuple $F_i(507 + 857k)$, whose admissibility is verified in [1], infinitely many of them also have a positive ES solution. An earlier version uses 800 forms; it has the explicit threshold $q > (Q + 16)^{393600}$ and the positive point $a = 10$, $h = 33$, $e = 1277$ on $n \equiv 1248 \pmod{1277}$.

Remark 4.12 (on Theorem 4.11). The proof in [1] combines a finite certificate with eventual arguments. The certificate is a packet of 159 constant nodes (84 offsets a and 75 rows h) and 315,642 signed divisor monomials, over a modulus of 287 digits with 91 support primes. The earlier 800-form packet had 302 nodes. The eventual arguments use polynomial division, with one common threshold. They also use “one-prime” fibre lemmas, which exhaust the fibre of an affine anchor whose auxiliary partner is not assumed prime. This is why only the primary forms $t + a$ and $ph - t$ need to be prime. The construction also uses the cases $z < 0$ and $z \geq p$ of Theorem 2.10 (for anchors that are eventually negative or $\geq p$) and Type II rigidity. Both predate the separation of the two projects. It does not use the Vieta–Jacobi case $2t < z < p$.

Our verification status. We re-ran the companion’s finite checker on a copy of its data; it reports VERIFIED_PRIMARY_PACKET_EVENTUAL for the packet

with SHA-256 62a88612...c2f2. We checked the positive solution in (2) symbolically. We have *not* independently re-derived the eventual closure argument of [1]. The proof of Theorem 4.11 is the companion's.

Theorem 4.11 is stronger than Theorem 4.13 below in three ways. It is linear: Dickson's conjecture, where one form is Dirichlet's theorem, against quadratic prime values, which are not known even for a single polynomial. It is smaller: 159 forms against 6402, which also improves the Bateman–Horn density. And it shows that ES holds at the primes of the subprogression $n \equiv 507 \pmod{857}$.

Timeline. The companion's whole-component result first appeared in an 800-form version. That version precedes Theorem 4.13. The reduction to 159 forms came after Theorem 4.13. The two certificates were obtained independently.

Theorem 4.13 (CONDITIONAL on H; certificate CERTIFIED). *There is an explicit frame $\mathfrak{F}$ with the following data: $|\Lambda| = 2036$, the largest prime in Λ about $3.37 \cdot 10^{16}$, $E_\ell \leq 17$ and $M \approx 10^{12088}$. The set S_e consists of 6402 polynomials (5009 linear, 1393 quadratic, $\sum \deg = 7795$), and S_a of 7119 further ones. There is also a connected set $\mathcal{V}$ of 7883 formal vertices satisfying (E1)–(E4). Consequently, under Hypothesis H for the 6402 polynomials f_g , $g \in S_e$, infinitely many primes $p = 24q + 1$ (q prime) have a seed component that equals the value set of $\mathcal{V}$ and contains no positive vertex. Under Bateman–Horn there are $\gg N(\log N)^{-6402}$ such $p \leq N$.*

Proof. Apply Theorem 4.7 to the certificate described below. $\square$

The certificate. The file `data/formal_closure/certificate.json.gz` contains $q_0 \bmod \ell^{E_\ell}$ for every $\ell \in \Lambda$; the 13521 polynomials of S , with their C_g and entry/auxiliary flags; and the 7883 formal vertices, as triples $(c, \{(g, e_g)\})$. The 7883 vertices have 9961 non-dead denominators (8412 p -divisible and 1549 p -free anchors with $0 < Z \leq 12X$) and 5726 dead ones. The certificate was found by a fixed-point iteration. A closure engine expands formal fibres from the seed, and the class q_0 is adjusted at primes where the candidate decisions (B) or the roots of S would otherwise be non-generic. It stabilises after nine rounds. The final closure uses no genericity assumption: every prime of every constant c_r is in Λ .

What is checked (CERTIFIED). An exact verifier checks the following:

- every $g \in S$ is primitive, irreducible and has positive leading coefficient, and C_g is correct;
- every vertex satisfies the identity in $\mathbb{Q}[X]$, has formal-integer entries, and has an entry with negative constant (E4);
- the dead/anchor/ p -divisible classification;
- every non-dead fibre, recomputed and compared with $\mathcal{V}$ (E2), and the precision condition (F2);
- condition (E3) for all 1271 primes $\ell \notin \Lambda$ with $\ell \leq 17631$;
- connectivity of $\mathcal{V}$.

The verifier `formal2_verify.py` reported OK with 0 mismatches. It was re-run by a second session, together with `formal2_verify_extra.py`. A hostile reviewer then wrote a fibre engine from scratch that shares no code with the generator. It recomputed all 9961 fibres: 533,011,471 candidates passed (A), and all fibres equal those of $\mathcal{V}$, with no prime of a c_r outside Λ and no auxiliary factor outside S . The reviewer also brute-forced 16 fibres at actual q chosen to make the relevant values prime, and all 16 agree with the formal prediction. For this paper, `formal2_verify_extra.py`, the reviewer’s global check and the reviewer’s fibre engine were run once more. All three report OK, and the fibre engine again finds 9961/9961 equal fibres.

Replay.

```
F=data/formal_closure
uv run --with python-flint python scripts/formal2_verify.py \
    $F/lam_final.json $F/closure_final.json.gz
uv run --with python-flint python scripts/formal2_verify_extra.py \
    $F/lam_final.json $F/closure_final.json.gz
export PYTHONPATH=scripts
uv run --with python-flint python scripts/review_fc_global.py
uv run --with python-flint python scripts/review_fc_fibres.py \
    --all --jobs 4 --out /tmp/review_fc_fibres_all.json
```

The first command took 2 h 50 min on 14 cores; the second takes 4 minutes and the third under one minute. The from-scratch fibre engine took 12 minutes on 8 cores.

Remark 4.14 (characters in Theorem 4.13). The class q_0 of Theorem 4.13 is *not* a square class: $24q_0 + 1$ is a non-residue modulo 870 of the 2036 primes of Λ . These are primes that were adjoined to fix the decisions (B) and the roots of S . So (E4) is not an instance of Proposition 4.6; it is verified directly. Theorem 4.7 itself never uses characters. Only 6402 of the 13521 polynomials are required to be prime: the auxiliary factors of $4Z - P$ enter only through (A) and (B), which use $n_h(q) \rightarrow \infty$ and Λ -freeness (Lemma 4.5).

Evidence 4.15 (actual q in the certificate’s class). Take 40 primes $q \in [10^9, 10^{11}]$ with $p = 24q + 1$ prime and $q \equiv q_0$ modulo $2^4 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$. For each, an exact layered BFS gives the following.

- The first layer (80 vertices) is always entirely formal.
- Every component has a positive vertex at distance 2.
- On every escape path, the path leaves the formal set at a formal denominator one of whose formal primes is not an actual prime. In 31 cases $g(q)/C_g$ is not even an integer; in 9 it is composite.

So actual components follow the formal one exactly as far as the admissibility conditions hold. This illustrates the mechanism; it is not a test of H.

5. FURTHER OBSTRUCTIONS

Each result below rules out a natural route to Conjecture 1.2 or to a pointwise proof.

5.1. Large denominators and Type II buckets.

Lemma 5.1 (PROVED). *The vertex $(2t, 2t, -pt)$ is the only Type I vertex whose two p -free denominators both lie in $[1, 2t]$. Consequently every other Type I vertex is the unique vertex at its second p -free denominator.*

Proof. Let (x, y, pm) have $x, y \in [1, 2t]$. Then $1/x + 1/y \geq 1/t > 4/p$, so $m < 0$. If $x, y \leq t$, then $1/x + 1/y \geq 8/(p-1) > 5/p \geq 4/p + 1/(p|m|)$, which is impossible. So say $x = t + a$ with $1 \leq a \leq t$, and chart the vertex from x with $h \leq 0$ (Lemma 2.7). Then $y = (t+a)(p|h|+t)/((4a-1)|h|+a)$. For $h = 0$, $y = t + t^2/a \leq 2t$ forces $a = t$, which gives $(2t, 2t, -pt)$. For $|h| \geq 1$ we get $y > 2t$, because $(t+a)(4t+1) - 2t(4a-1) = 4t(t-a) + 3t + a > 0$ and $t(t+a) \geq 2ta$. The last sentence follows from Theorem 2.10. $\square$

By Theorem 2.10, Lemma 5.1, Corollary 2.6 and Corollary 2.11, every edge of G_p passes through a p -free denominator in $[1, 2t]$ or through a p -divisible one. Type II buckets never mix signs and have at most two vertices. Every sign change happens at an anchor in $(t, 2t]$ or at a Type I bucket with $h \geq 1$. *What this kills:* long hidden escape routes through large denominators, and any argument that changes sign inside a Type II bucket. It is also what makes the formal closures of §4 finite.

5.2. Dead hubs and sterile components larger than the seed component. Numerically, for typical p the seed component is the largest component (§6). This suggested a *size conjecture*: every component with more than $K(p) = O(\log p)$ vertices contains a positive vertex. Combined with “the seed component is large”, it would have implied Conjecture 1.2.

Lemma 5.2 (negative quadrant; PROVED). *A vertex containing a p -free denominator $x \in [1, t]$, or a Type I denominator $p(ph - t)$ with $h \leq 0$, is nonpositive.*

Proof. $4/p - 1/x < 0$ in the first case, and $ph - t < 0$ in the second. $\square$

Lemma 5.3 (dead-hub fibre; PROVED). *Let $k \geq 2$, $\mathcal{M} = 4k + 1$, and suppose that $x = t - k \geq 1$ has all prime factors $\equiv 1 \pmod{\mathcal{M}}$. Then the vertices containing x are exactly*

$$V_d = \left(x, -\frac{p(x-d)}{\mathcal{M}}, \frac{px(x-d)}{\mathcal{M}d} \right), \quad d \mid x^2, \ 0 < d < x.$$

There are $(\tau(x^2) - 1)/2$ of them; all are of Type II and nonpositive.

Proof. Here $4/p - 1/x = -\mathcal{M}/(px)$ with $\gcd(\mathcal{M}, px) = 1$, so Lemma 2.1 applies with $r = -\mathcal{M}$ and $s = px$. The divisors are $D = \pm p^j d$ with $d \mid x^2$ and $j \in \{0, 1, 2\}$, and the condition is $\mathcal{M} \mid px + D$. Modulo $\mathcal{M}$ we have $d \equiv x \equiv 1$ and $p = 4x + \mathcal{M} \equiv 4$, so the condition reads $\pm 4^j \equiv -4$. For $j = 0$ it needs $\mathcal{M} \mid 3$ or $\mathcal{M} \mid 5$; for $j = 2$ it needs $\mathcal{M} \mid 12$ or $\mathcal{M} \mid 20$; for $j = 1$ the sign $+$ needs $\mathcal{M} \mid 8$. Since $\mathcal{M} \geq 9$ is odd, only $D = -pd$ survives. It gives the entries $-p(x-d)/\mathcal{M}$ and $px(x-d)/(\mathcal{M}d)$. The value $d = x$ is the excluded $D = -s$, and $d \mapsto x^2/d$ swaps the two p -divisible entries. $\square$

Lemma 5.4 (descent; PROVED). *In the situation of Lemma 5.3, let $d \mid x$ with $w = x/d > 1$ and $w \equiv 1 \pmod{4}$. Then*

$$D_d = \left(x - d, -\frac{p(x-d)}{\mathcal{M}}, -\frac{p(w-1)}{4} \right)$$

is a vertex sharing $-p(x-d)/\mathcal{M}$ with V_d , and the bucket of this denominator is exactly $\{V_d, D_d\}$. Its p -free entry $x-d = d(w-1) < t$ is again a negative-quadrant anchor.

Proof. Integrality follows from $\mathcal{M} \mid x-d$ (all prime factors of x are $\equiv 1 \pmod{\mathcal{M}}$) and $4 \mid w-1$. The identity is $1/(d(w-1)) - \mathcal{M}/(pd(w-1)) - 4/(p(w-1)) = (p - \mathcal{M} - 4d)/(pd(w-1)) = 4/p$, using $p = 4dw + \mathcal{M}$. The bucket is of Type II, so it has at most two vertices (Lemma 2.9). $\square$

Guarded hubs (EVIDENCE for the closure). Take x squarefree, $x = q_0 \ell_1 \cdots \ell_r$. The primes ℓ_i are $\equiv 1 \pmod{4\mathcal{M}}$, and the guard $q_0 \sim 10^4$ is $\equiv 1 \pmod{\mathcal{M}}$ and $\equiv 3 \pmod{4}$. Then the hub fibre has $(3^{r+1} - 1)/2$ vertices (Lemma 5.3). The divisors $w > 1$ of x with $w \equiv 1 \pmod{4}$ are exactly the $2^r - 1$ divisors $w > 1$ of x/q_0 , so Lemma 5.4 gives exactly $2^r - 1$ descents, all with $d \geq q_0$. In most computed instances no further vertex appears (§3.4 of the source notes: 122 of 126 guarded instances were sterile). The component then has $(3^{r+1} - 1)/2 + 2^r - 1$ vertices. That closure is verified instance by instance, not proved.

Evidence 5.5 (CERTIFIED sterile components). Each component below was exhausted. An independent checker verified every vertex identity, nonpositivity, connectivity and closure: the full fibre of every denominator lies in the set. Anchors and almost all Type I buckets were recomputed by two independent methods; Type II and outer fibres use the factorisation-free rules of Lemma 2.9 and Theorem 2.10. A hostile review re-ran the checker, together with its own fibre code, on four certificates.

	p	sterile component	$\ln p$	size/ $\ln p$
165426666927637		398	32.7	12
1960717994383909 (random p)		339	35.2	9.6
475041241432047253		3469	40.7	85
274159709010072908384347957		30035	61.9	485

At $p = 274159709010072908384347957$ the seed component was also exhausted. It has 10155 vertices, 2811 of them positive, while the sterile component has 30035. In 13 of 14 comparable dead-hub primes the sterile component was larger than the seed component.

What this kills: any argument that forces positivity in the seed component because it is large. At the prime above, a larger component is sterile. The same applies to any “giant component” statement in the style of Bourgain–Gamburd–Sarnak [2] with the seed as the giant. Variants of the same construction give certified sterile components with 243 distinct Type I buckets,

and with 146 failing positive-capable denominators. The asymptotic $O(\log p)$ form is not formally refuted, since sterility of an infinite family is not proved.

Lemma 5.6 (sign flip; PROVED). *Let $x = t + a$ with $a \geq 1$ and $p \nmid x$, and suppose that x has a prime factor $\ell \equiv -1 \pmod{4a-1}$. If x occurs in some vertex, then it occurs in a positive one. Likewise, if $h \geq 1$ and $m = ph - t$ has a prime factor $\ell \equiv -1 \pmod{4h-1}$, then the Type I bucket of pm is empty or contains a positive vertex.*

Proof. Both cases use Lemma 2.1 with $r, s > 0$ and $\gcd(r, s) = 1$. For the anchor, $r = q := 4a - 1$ and $s = px$; for the bucket, $r = H := 4h - 1$ and $s = m$. A vertex of the fibre is positive if and only if its divisor D is positive. Indeed, $D > 0$ makes both entries positive. If $D < 0$, both factors $ry - s$ and $rw - s$ are negative, and $y, w > 0$ would give $r/s = 1/y + 1/w > 2r/s$. Now take a vertex of the fibre with $D < 0$. Write $D = -p^j d$ with $d \mid x^2$ (anchor), or $D = -d$ with $d \mid m^2$ (bucket). Let $u = x$ (anchor) or $u = m$ (bucket), so $\ell \mid u$ and $d \mid u^2$. Replace d by $d' = \ell d$ if $v_\ell(d) < 2v_\ell(u)$, and by d/ℓ otherwise; in both cases $d' \mid u^2$. Since $\ell \equiv \ell^{-1} \equiv -1 \pmod{r}$, we have $d' \equiv -d \pmod{r}$. So $D' = p^j d'$ (respectively $D' = d'$) is a positive divisor of s^2 with $D' \equiv D \pmod{r}$, and it gives a positive vertex. $\square$

So in a sterile component every positive-capable anchor or bucket avoids prime factors $\equiv -1$ modulo its own modulus. This is a classical divisor-class restriction, not a bound on size. The certified dead hubs contain few positive-capable denominators, all of large modulus. The seed component, by contrast, always contains the cheap tests $h \mid t^2$. Any replacement statement must weigh tests by their moduli, and this leads back to the classical divisor-class problem.

5.3. Parity and windmills. Zagier's one-sentence proof [18] suggests looking for a finite set with two involutions.

Lemma 5.7 (PROVED). *Let σ_1, σ_2 be involutions of a finite set S with $|\text{Fix } \sigma_1|$ odd, and let $\Phi : \text{Fix } \sigma_2 \rightarrow \{\text{positive vertices}\}$. Then $\sum_v |\Phi^{-1}(v)|$ is odd. So any such argument exhibits an explicit weight on positive solutions whose total is odd for every p .*

The plain count is not such a weight: the number of positive solutions is even for $p = 97$ (it is 8) and for $p = 193$ (it is 6).

Proposition 5.8 (PROVED).

- (1) (Repeated denominators.) *The only vertices with a repeated denominator are σ and $(2t, 2t, -pt)$. So coordinate permutations have exactly two non-free orbits, and both are nonpositive.*
- (2) (Fibre parity.) *Let $x > p/4$ be p -free, $r = 4x - p$, $s = px$. Let $N^\pm$ be the number of positive $D \mid s^2$ with $D \equiv \mp s \pmod{r}$. Then N^+ is even and N^- is odd. The fibre has $N^+/2$ positive vertices and $(N^- - 1)/2$ vertices with exactly one negative entry. So the odd count in a fibre always certifies mixed-sign vertices.*

- (3) (Affine symmetries.) *In the classical four-parameter model, positive Type II solutions are $(x, y, z) = (abc, pkbc, pkac)$ with $k(4abc - p) - a - b = 0$. Positive Type I solutions are $(pabc, kbc, kac)$ with $k(4abc - 1) - p(a + b) = 0$. Up to renaming $(c \mapsto d_{\text{ET}}, k \mapsto c_{\text{ET}})$ these are [8, (2.3), (2.15), (2.22), Prop. 2.2]. The maps $v \mapsto Av + \beta$ in $\text{AGL}_4(\mathbb{Z})$ that multiply the Type II polynomial by a nonzero constant form a group of order 4. The same holds for the Type I polynomial. On positive points only the swap $a \leftrightarrow b$ survives.*
- (4) (The natural odd set.) *The set of ordered positive solutions of $1/x + 1/y + 2/z = 4/p$ is finite and odd, but its part with z even has exactly $6f(p)$ elements, where $f(p)$ is the number of unordered positive solutions of (1) with $n = p$.*

Proof. (1) If $2/x + 1/z = 4/p$ with $p \nmid x$, then $z = px/(2(2x - p))$ and $2x - p = \pm 1$, so $x = 2t$ and $z = -pt$. If the repeated entry is pm , then $x = pm/(2(2m - 1))$ forces $m = -2t$ and $x = t$. (2) Complementation $D \mapsto s^2/D$ preserves both classes. Its unique positive fixed point $D = s$ lies in the class $+s$, not $-s$, since $r \nmid 2s$. Positive D in the class $-s$ give positive vertices; negative divisors give mixed ones, or $y = 0$ when $D = -s$. (3) The quartic part $4abck$ forces a monomial linear part with entries ± 1 . The absence of cubic terms forces the translation to vanish. The linear part $-pk - a - b$ forces $k \mapsto \pm k$ (as $p \neq \pm 1$), $\{a, b\}$ to be stable, and $c \mapsto c$. (4) The swap of x, y fixes exactly the three solutions of $1/x + 1/z = 2/p$, all with z odd. On z even, $(x, y, z) \mapsto \{x, y, z/2\}$ is 6-to-1 onto ES solutions by (1). $\square$

What this kills. Fibre-level parity (2) and coordinate permutations (1) give odd counts only of nonpositive objects. No integral affine windmill preserves the equation identically (3); unlike Zagier's quadratic form $x^2 + 4yz$, the four-parameter surface is multilinear and has no Vieta involutions. Elsholtz [7, §3] notes the same fragility of windmill partitions for general forms. A parity proof would need a new explicit odd weight (Lemma 5.7). EVIDENCE: among the 732 primes $p \equiv 1 \pmod{24}$ below $6 \cdot 10^4$, about 7000 candidate weights were tested, and none generalised beyond chance. Pieces that preserve only the finite point set are not excluded by (3).

6. NUMERICAL EVIDENCE

Everything in this section is EVIDENCE. Finite computations do not prove Conjecture 1.2, and by Theorems 4.9 and 4.13 the patterns are conditionally misleading.

Evidence 6.1 (seed distance below 10^{12}). Every prime $p \equiv 1 \pmod{4}$ with $p < 10^{12}$ has $\text{dist}(p) \leq 3$; exactly 44197 of them have $\text{dist}(p) = 3$, and no prime with $\text{dist}(p) \geq 4$ is known. By Corollary 3.4 only Mordell's six classes need testing. Their primes below 10^{12} were tested exhaustively by the exact

criterion (6), and every survivor was re-checked through the families (A), (B) of Theorem 3.1.

range	primes tested	dist = 3	dist ≥ 4
$p < 10^8$	179468	70	0
$10^8 \leq p < 10^{10}$	14036239	1306	0
$10^{10} \leq p < 10^{11}$	114455512	6196	0
$10^{11} \leq p < 10^{12}$	1046544177	36625	0

Independently, a complete BFS covered all $p \equiv 1 \pmod{4}$ below $3 \cdot 10^5$; only 297049 has dist = 3. Among the sparse inputs $p = 24q + 1$ with q prime and $q \leq 10^{13}$ (restricted to Mordell’s classes above $q = 10^{11}$, which loses nothing by Lemma 3.3), all 1113907 that fail (6) escape at depth three. All ten distance-3 primes below $5 \cdot 10^6$ escape through family (A), and so do all 5304 survivors with $10^8 < q \leq 10^{10}$; (B) also succeeds for about half of them, but never alone.

Evidence 6.2 (why nothing deeper is seen). Among the survivors of (6) with $10^8 < q \leq 10^{10}$, the number of distinct productive (A)/(B) anchors has mean 14.7 and minimum 2. The mean grows with p (17.3 near $q \approx 5 \cdot 10^{12}$), because the shifts $pc + t$ acquire more divisors. So within reach “distance ≤ 3 ” will keep looking true. Theorem 4.9 produces deep primes only from simultaneous primality of the family $S_e^{(3)}$, whose Bateman–Horn density is far beyond reach. The two pictures are consistent.

Evidence 6.3 (component sizes). Complete enumeration was carried out for all $p \equiv 1 \pmod{4}$ below $2 \cdot 10^6$, with samples up to 2^{31} . For typical p the seed component is the unique giant; its share of all vertices declines from 0.58 to 0.36. The mean per-prime maximal sterile size fits $(\log_2 p)^2/50$. Nearly every large sterile component is a hub around a negative-quadrant denominator (Lemma 5.2), which is exactly the mechanism amplified in Lemma 5.3.

7. WHAT A POINTWISE PROOF MUST DO

The results above locate the difficulty precisely.

- (1) *Positivity is a character event.* By Theorem 2.5 a vertex is positive exactly when its equal-valuation pair has opposite quadratic characters modulo p . This is the only Brauer–Manin invariant [4, Thm. 1.6], and it gives no obstruction to existence [4, Thm. 1.1]. It cannot create a sign change; it only detects one.
- (2) *Sign changes need fresh non-residues.* By Lemma 3.2, an exit requires a prime factor ℓ of a new shift with $\left(\frac{\ell}{p}\right) = -1$. Everything built from the support of t is a residue (a “principal-genus trap”).
- (3) *Formal arguments cannot supply them.* If every number met factors formally in a square class, every prime met is a residue (Proposition 4.6). Under H this happens for infinitely many primes, for every bounded-radius exploration from the seed (Theorem 4.9). This is the

Elsholtz–Tao odd-square principle in graph form. Beyond bounded radius, the whole seed component is closed by explicit finite certificates: under H in Theorem 4.13, and, as reported, under Dickson in Theorem 4.11. Their nonpositivity is checked directly; Theorem 4.13 is not a square-class instance (Remark 4.14).

- (4) *The global invariants we tested do not help.* Component size (§5.2), the parity constructions of §5.3, and quadratic or quartic colourings gave no invariant that forces positivity. Sterile components can be larger than the seed component. This is a record of failed attempts, not an impossibility theorem.

A pointwise proof of ES must therefore use information that a formally generic prime does not have. It must control the *actual* factorisation of specific shifted integers $t + a$, $ph - t$, $\dots$, in the required classes, for every p . This could come from the size of p relative to the numbers met, from analytic input about prime factors of shifted primes, or from a structure invisible to formal polynomial families. For the seed graph this is exactly the problem that the exceptional-set machinery controls only on average: Vaughan’s bound $E(N) \ll N \exp(-c(\log N)^{2/3})$ [16] and its refinements. Conditionally on H , Conjecture 1.2 is false (Theorem 4.13). By the reported Theorem 4.11, under Dickson for the restricted tuple, reachability fails even at primes where ES holds. So universal seed reachability is not a route to ES. Further work should return to the exceptional-set problem, or find genuinely non-formal input.

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Theorem 4.11 and its proof are due to the companion computation [1]. It reached the conditional disproof independently and first, in an 800-form version that was later reduced to 159 forms (after Theorem 4.13), and in the stronger linear form.

APPENDIX A. REPLAY

All commands are run from the repository root, and PY abbreviates

```
PYTHONPATH=scripts uv run --with python-flint python
```

We ran everything under `ulimit -v 8000000` (or the stated memory) with `timeout`, on at most 4 cores. Times are wall-clock.

```
# Section 2-3: base lemmas, Theorem 3.1, Lemma 3.3 (each < 1 min)
PY scripts/pointwise_seed_check.py          # 10 s, prints OK
PY scripts/pointwise_fibres_check.py         # 6 s, prints OK
PY scripts/pointwise_incidence.py 297049     # 3 s, 1143 vert.
PY scripts/depth3_check.py                   # 5 s, prints OK
PY scripts/depth3_validate.py 13 30000 --jobs 4 # 34 s, 0 mismatches
PY scripts/depth5_branches.py 13 3000000     # 16 s
```

```
# Evidence 6.1: survey of Mordell’s classes (P0 P1 THREADS)
```

```

g++ -O3 -march=native -std=c++17 -pthread \
  -o /tmp/d3allp scripts/depth3_allp.cpp
/tmp/d3allp 13 100000000 4 > /tmp/allp.txt # 2 s, 70 candidates
PY scripts/depth3_batch.py /tmp/allp.txt --jobs 4 # 2 s, all dist 3
# sparse t=6q: (S Q0 Q1 THREADS [hard]), then the same batch check
g++ -O3 -march=native -std=c++17 -pthread \
  -o /tmp/d3sieve scripts/depth3_sieve.cpp
/tmp/d3sieve 6 1 100000000 4 hard > /tmp/surv.txt # 217 survivors
PY scripts/depth3_batch.py /tmp/surv.txt --jobs 4

# Theorem 4.13: see Section 4.7 for the four verification commands
# Evidence 4.15: actual q in the certificate's class
PY scripts/formal2_realq.py data/formal_closure/lam_final.json \
  data/formal_closure/closure_final.json.gz \
  --mod-bound 13 --power-cap 30 --n 40 # 6 s

# Evidence 5.5: one certificate per run
PY scripts/sterile_certificate_check.py \
  data/sterile/certs/FILE.json.gz
# p165426666927637, p1960717994383909_random,
# p475041241432047253, p2192882958603411108997: about 10 s each
# p433...441_tests, p691...041_typeI: about 40 s each
# p274159709010072908384347957: 4.5 min, 1.6 GB resident
# p274159709010072908384347957_seed: 23 min

# Section 5.3: windmill scan (T is mandatory)
g++ -O2 -std=c++17 -o /tmp/wm_single scripts/windmill_singleton.cpp
/tmp/wm_single 2500000 # 1 s, 2939478 candidates, 0 hits

```

Files of Theorem 4.13. The primary certificate is `data/formal_closure/certificate.json.gz`. It holds $q_0 \bmod \ell^{E_\ell}$, S with the C_g and entry/aux flags, and the 7883 vertices. The verifiers `formal2_verify.py` and `formal2_verify_extra.py` read the equivalent pair `lam_final.json` and `closure_final.json.gz`. The first gives the residues of Λ to precision $\geq E_\ell$, from which the model point $q_t \equiv q_0 \pmod{\ell^{E_\ell}}$ is rebuilt. The second gives S , the C_g , the vertices and the precision data. Its S , C_g and vertex set are identical to the certificate's. The reviewer's scripts read all three files and check that they agree. The fixed-point construction (`scripts/formal2_iter.py`) is driven by subcommands; its full replay sequence is in `FORMAL_CLOSURE.md`, §6. The companion's checker for Theorem 4.11 is `scripts/pointwise_primary_packet_check.py` in [1] (about 1 minute).

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